Reading group notes: linear attention

Reading group of the Ashbury machine learning lab

1 Background

Most prior methods treat the causal mask as fixed and tune the chunk-wise kernel around it. Despite steady progress, the interaction between the associative recall and the key dimension is still poorly understood. This work studies linear attention, a method that rethinks the outer-product write. Earlier linear attention methods improved the outer-product write but left the state size unchanged.

Our analysis covers models up to 1.3B parameters, and larger models may behave differently. We expect the read-out projection to interact with the associative recall at larger scales, which we leave to future work. A limitation of linear attention is that the feature map must be tuned separately for each model size. The key dimension also makes linear attention easier to inspect, since its behaviour can be read off per token.

We argue that the feature map deserves as much attention as the causal mask. Practitioners report that the chunk-wise kernel is the first component to fail on long inputs. Most prior methods treat the recurrent state as fixed and tune the decay gate around it. A practical system has to balance the cost of the chunk-wise kernel against the quality of the causal mask.

2 Methods we read

During inference linear attention caches the causal mask, so each new token costs a constant amount of work. Each block of linear attention applies the feature map after a short convolution of width 2. A coefficient controls how strongly the key dimension depends on the associative recall; we set it to 0.4 throughout. To keep training stable, linear attention normalises the state size before it is combined with the associative recall.

A coefficient controls how strongly the outer-product write depends on the chunk-wise kernel; we set it to 0.2 throughout. We implement the outer-product write as a fused kernel, which avoids writing intermediate results to memory. The chunk-wise kernel is initialised so that linear attention starts training close to softmax attention. Linear attention computes the causal mask once per layer and reuses it for every head.

During inference linear attention caches the chunk-wise kernel, so each new token costs a constant amount of work. A coefficient controls how strongly the causal mask depends on the read-out projection; we set it to 0.7 throughout. To keep training stable, linear attention normalises the outer-product write before it is combined with the decay gate. Each block of linear attention applies the chunk-wise kernel after a short convolution of width 5.

3 Results we discussed

On WikiText, linear attention improves over gated linear attention by 3.0 points. Ablating the read-out projection lowers the MQAR score by 2.1 points, the largest drop among all components. Throughput is measured with a batch size of 256 on a single accelerator. Variance across seeds stays below 0.8 points for every configuration we report.

Ablating the decay gate lowers the LAMBADA score by 4.9 points, the largest drop among all components. On LAMBADA, linear attention improves over softmax attention by 6.1 points. Linear attention scores 60.6% on LAMBADA, compared with 92.6% for gated linear attention. With the key dimension frozen, linear attention still recovers most of its improvement on MQAR.

Linear attention scores 93.6% on WikiText, compared with 91.9% for gated linear attention. With the read-out projection frozen, linear attention still recovers most of its improvement on LAMBADA. Ablating the feature map lowers the LAMBADA score by 4.8 points, the largest drop among all components. The gains of linear attention grow with the input length, as the key dimension becomes the bottleneck.

4 Open questions

The results suggest that the chunk-wise kernel is the main lever for quality at a fixed budget. We expect the key dimension to interact with the feature map at larger scales, which we leave to future work. The recurrent state also makes linear attention easier to inspect, since its behaviour can be read off per token. A limitation of linear attention is that the associative recall must be tuned separately for each model size.

The results suggest that the feature map is the main lever for quality at a fixed budget. We expect the recurrent state to interact with the chunk-wise kernel at larger scales, which we leave to future work. Our analysis covers models up to 1.3B parameters, and larger models may behave differently. The feature map also makes linear attention easier to inspect, since its behaviour can be read off per token.